

# Numerical Analysis

## Lab 2

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# Rootfinding for nonlinear equations

This lab deals with the numerical approximation of the zeros of a real valued function of one variable, that is

given  $f : \mathcal{I} = (a, b) \subset \mathbb{R} \rightarrow \mathbb{R}$ , find  $\alpha \in \mathbb{R}$  such that  $f(\alpha) = 0$ .

Methods for the numerical approximations of a zero of  $f$  are usually iterative; the aim is to generate a sequence of values  $x^{(k)}$  such that

$$\lim_{k \rightarrow \infty} x^{(k)} = \alpha$$

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## Theorem 2.1 (Bolzano's theorem)

Let  $f \in \mathcal{C}^0([a, b])$  with  $f(a)f(b) < 0$ , then  $\exists \alpha \in (a, b)$  s.t.  $f(\alpha) = 0$ . Moreover, if  $f$  is monotone in  $[a, b]$  (e.g.,  $f \in \mathcal{C}^1([a, b])$  and  $f'$  has constant sign), then  $\alpha$  is unique.

## Method 2.1 (Bisection method)

Let  $x_{left}^{(0)} = a$ ,  $x_{right}^{(0)} = b$ ; at step  $k \in \mathbb{N}$  compute the midpoint  $\bar{x}^{(k)} = (x_{left}^{(k)} + x_{right}^{(k)})/2$  and  $f(\bar{x}^{(k)})$ :

- if  $f(x_{left}^{(k)})f(\bar{x}^{(k)}) < 0$ , set  $x_{left}^{(k+1)} = x_{left}^{(k)}$  and  $x_{right}^{(k+1)} = \bar{x}^{(k)}$ ;
- else set  $x_{left}^{(k+1)} = \bar{x}^{(k)}$  and  $x_{right}^{(k+1)} = x_{right}^{(k)}$ .

and stop when the number of iteration  $n = N_{Max} = \lceil \log_2 \left( \frac{b-a}{\text{tol}} \right) \rceil = \left\lceil \frac{\log \frac{b-a}{\text{tol}}}{\log 2} \right\rceil$

## Comments

- Global convergence, but not monotone decreasing of the error
- slow convergence
- Need to provide upper and lower bounds

## Exercise 2.1

Consider the following function

$$f(x) = x^3 - (2 + e)x^2 + (2e + 1)x + (1 - e) - \cosh(x - 1), x \in [0.5, 5.5].$$

- a Plot the function  $f$  and determine two intervals that contain its roots (use the Matlab command `grid on`).
- b For which roots the bisection method can be used? Compute the number of needed iterations for the bisection method to converge with a tolerance of  $10^{-3}$ , when the interval  $[3, 5]$  is chosen as starting interval.
- c Implement the bisection method:  

```
function [x,x_iter]=bisection(f,a,b,tol)
```

where  $x$  is the solution,  $x\_iter$  is the vector of the approximations at each iteration,  $f$  is the function, defined as handle function,  $a, b$  are the end points of the interval,  $tol$  is the required tolerance.
- d Employ the bisection method to approximate the roots with a tolerance equal to  $10^{-3}$ .
- e Plot the evolution of the error with the iterations.

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## Method 2.2 (Newton method)

Newton method consists in approximating the zero of  $f(x)$  with the sequence

$$x^{(k+1)} = x^{(k)} - \frac{f(x^{(k)})}{f'(x^{(k)})}, k \geq 0, f'(x^{(k)}) \neq 0$$



## Theorem 2.2

If  $f \in \mathcal{C}^2([a, b])$  and  $f'(x) \neq 0$  in an open interval containing  $\alpha$ , then  $\exists \delta > 0$  s.t.  $\forall x^{(0)} : |x^{(0)} - \alpha| < \delta$  the Newton method converges **quadratically** to  $\alpha$ . Moreover, if in addition  $f(a)f(b) < 0$  and  $f''(x)$  has constant sign in  $[a, b]$ , the method converges for any  $x^{(0)} \in [a, b]$  s.t.  $f(x^{(0)})f''(x^{(0)}) > 0$ .

## Comments

- conditional convergence (dependence on the initial guess  $x^{(0)}$ )
- $\alpha$  simple zero ( $f'(\alpha) \neq 0$ ): quadratic conditional convergence
- $\alpha$  zero of multiplicity  $m$  ( $f^{(i)}(\alpha) = 0, i = 0, \dots, m-1, f^{(m)}(\alpha) \neq 0$ ): linear conditional convergence

**Convergence order of a scheme  $\{x^{(k)}\}_{k \in \mathbb{N}}$  approximating a zero  $\alpha$**

$$p = \lim_{k \rightarrow \infty} \frac{\log \frac{e^{(k+1)}}{e^{(k)}}}{\log \frac{e^{(k)}}{e^{(k-1)}}},$$

with  $e^{(k)} = |x^{(k)} - \alpha|$  error at the  $k$ -th iteration.

## Exercise 2.2

Consider the following function in the interval  $[-0.5, 1.5]$

$$f(x) = \sin(x)(1 - x)^2.$$

- Plot  $f$  in order to find some intervals containing the roots.
- Implement the Newton method by using a stopping criterion based on the error estimator  $|x^k - x^{k-1}|$ . The signature of the function is:  
`function [x,x_iter]=newton(f,df,x0,tol,Nmax)`  
where  $x$  is the approximate,  $x\_iter$  is the vector of the approximations at each iteration,  $f$ ,  $df$  are the function and its first derivative, defined as handle functions,  $x_0$  is the initial guess,  $tol$  is the tolerance demanded by user and  $Nmax$  is the maximum number of allowed iterations.
- Use Newton method to find the roots with a tolerance equal to  $10^{-6}$ , by considering as initial guess  $x_0 = 0.3$  and  $x_0 = 0.5$ .
- Compute an estimate of the convergence rate. Is it the expected one?

## Exercise 2.3

Consider the following function in the interval  $[-2, 2]$

$$f(x) = \exp(x) - x.^2 - \sin(x) - 1;$$

- Plot  $f$  and its two first derivatives. How many roots does the function have? What is the multiplicity of each root?
- Find the roots of the function using the Newton method with a tolerance of  $10^{-6}$  and initial guess  $x^{(0)} = 0.1$  for the root with  $m > 1$  and  $x^{(0)} = 1.2$  for the simple root. Compute the error considering the exact solutions are  $x_1 = 0$  and  $x_2 = 1.2797$ .
- Compute an estimate of the convergence rate. Is it the expected one?

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## Theorem 2.3

If  $f \in \mathcal{C}^2([a, b])$  and  $f'(x) \neq 0$  in an open interval containing  $\xi$ , then  $\exists \delta > 0$  s.t.  $\forall x^{(0)} : |x^{(0)} - \xi| < \delta$  the Newton method converges quadratically to  $\xi$ .

The convergence is guaranteed only if the initial guess  $x^{(0)}$  is close enough to the root  $\xi$ , and for this reason the Newton method is a **locally** convergent method. A simple solution to overcome this issue consists in employing the bisection method to predict the initial guess  $x^{(0)}$ , as shown in the next exercise.

## Exercise 2.4

Consider the following function in the interval  $[-1, 6]$

$$f(x) = \arctan \left[ 7 \left( x - \frac{\pi}{2} \right) \right] + \sin \left[ \left( x - \frac{\pi}{2} \right)^3 \right].$$

- a Plot  $f$  in order to find an interval containing a root. What is the multiplicity of the root?
- b Use the Newton method to find the root with a tolerance of  $10^{-10}$  and initial guess  $x^{(0)} = 1.5$ . Compute the error.
- c Use the Newton method to find the root with a tolerance of  $10^{-10}$  and initial guess  $x^{(0)} = 4$ . Compute the error.
- d If possible, apply the bisection method on the interval  $[a, b] = [-1, 6]$  and tolerance  $\frac{b-a}{2^{30}}$ . Compute the error.
- e Write a function `bisection_newton.m` to find  $\xi$  using the Newton method starting from an initial guess obtained after few iterations of a bisection method. Test with  $[a, b] = [-1, 6]$ , 5 iterations of the bisection method and tolerance  $10^{-10}$  for the Newton method.

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## 4 Homework



## Homework 2.1 (Bisection method)

Consider the following equation

$$\cot(x) = \frac{x^2 - 1}{2x}.$$

- a Define three intervals containing the three smallest positive roots.
- b Find the biggest root among the ones of point 1, using bisection method with tolerance  $10^{-6}$ .
- c Plot the evolution of the error with the iterations.

## Homework 2.2 (Newton method)

- a Approximate the root  $x^*$  of  $\tan(x^*) = 2x^*$  in the interval  $(0, \frac{\pi}{2})$  up to a tolerance of  $10^{-15}$ .
- b Approximate the root of  $\sin(x)$  in the interval  $[-\frac{\pi}{2}, \frac{\pi}{2}]$  using Newton method and starting from the value  $x^{(0)} = x^*$ . What do you expect? What do you observe?

## Homework 2.3 (Newton method)

Consider the following function

$$f(x) = e^{ax} - 1 = 0, \quad a \neq 0.$$

For any value of the parameter  $a$  the unique solution is clearly  $\xi = 0$ .

- a Consider the case of  $a = 200$  and apply Newton method starting from  $x^{(0)} = 1$  with tolerance values of  $10^{-3}$  and  $10^{-12}$ .
- b Do the same for  $a = 10^{-3}$ .
- c For each case compare the absolute value of the residual with the error. Motivate the obtained results.

# Homework IV

Newton method requires the analytical expression of  $f'(x)$ . To avoid this limitation the derivative could be approximated.

## Secant method

Using 
$$f'(x^{(k)}) \simeq \frac{f(x^{(k)}) - f(x^{(k-1)})}{x^{(k)} - x^{(k-1)}}$$

$$x^{(k+1)} = x^{(k)} - \frac{f(x^{(k)})(x^{(k)} - x^{(k-1)})}{f(x^{(k)}) - f(x^{(k-1)})}$$

superlinear order

$$\rho = \frac{\sqrt{5}+1}{2} \simeq 1.61.$$

## Steffensen method

Using 
$$f'(x^{(k)}) \simeq \frac{f(x^{(k)} + f(x^{(k)})) - f(x^{(k)})}{f(x^{(k)})}$$

$$x^{(k+1)} = x^{(k)} - \frac{(f(x^{(k)}))^2}{f(x^{(k)} + f(x^{(k)})) - f(x^{(k)})}$$

second order  
as Newton  
method.

## Homework 2.4 (Secant method, Steffensen method)

Implement both the secant method and Steffensen method. Test them on a problem and compare the convergence rate with respect to Newton method.

## Homework 2.5 (Newton method)

The equation  $x^3 - x^2 - x + 1 = 0$  has a double root in  $\xi = 1$ . Given  $x_0 = 2$ , does Newton method converge to  $\xi$ ? How can you improve the method for approximating  $\xi$ ? Compare the results.

## Homework 2.6 (Bisection method)

Consider the following equation

$$e^{-(x-2)^2} = 1 - e^{x-4}.$$

- a Plot the function and determine the number of real solutions.
- b Apply the bisection method with tolerance  $10^{-3}$  with initial intervals  $I_1 = [0, 5]$ ,  $I_2 = [1, 6]$ ,  $I_3 = [1, 5]$  and  $I_4 = [-1, 5]$ . Can you find all the roots?

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